Evaluation runs: lock (a3-lock)

generated 2026-11-08 from brain-data 38584f536 2026-11-08 20:45:11 +0100

Every campaign run on this benchmark, one row per instance (variant, specification, communication model, size) and engine; the newest run of an instance wins. Expected verdicts (the rule file): fifo: expected conforms; control-lifo: expected conforms; wrong-round: expected violation; double-grant: expected violation; double-grant: expected conforms (one client); never-grant: expected violation; never-grant: expected conforms (one client). Blue rows are at the current pins (TraceForge 367cdc6, whose predecessor 9a0629e is equivalent for every engine but the join; merge join 367cdc6; JMC 3c43b2d2): a rerun gives the same result. Red rows are superseded (an older pin, or the in-memory join that the merge join replaces) and their rerun is pending. A planned instance without a measurement is not a row: it is listed under "Not run". G_{ref}, G_{cand} = the graphs of the reference and of the implementation (the stateless engine's own count of specification graphs explored, repeats included, is G_{ref}^{expl}); G_x = the implementation graphs no specification graph covers (violating pairs); T = wall seconds; M = peak resident MiB; tier = the run's time and memory limits; * marks a verdict other than the expected one; 🕓 = the time tier was reached; oom = the memory limit was reached.

stateless

stateless: pair control-lifo, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	sweeps	G_{ref}^{expl}	run
2,2	6	28	conforms	0.0	2	1 h, 250 GB	6	21	brain03, 2026-11-02
2,3	20	212	conforms	0.1	4	1 h, 250 GB	20	210	brain03, 2026-11-02
2,4	70	1676	conforms	0.2	2	1 h, 250 GB	75	2625	brain03, 2026-11-02
2,5	252	13604	conforms	0.9	2	1 h, 250 GB	516	65867	brain03, 2026-11-02
2,6	924	112380	conforms	9.4	4	1 h, 250 GB	2424	1192017	brain03, 2026-11-02
3,2	90	2850	conforms	0.3	2	1 h, 250 GB	98	4366	brain03, 2026-11-02
3,3	1680	324282	conforms	26.7	23	1 h, 250 GB	3214	2658596	brain03, 2026-11-02
3,4	34650		🕓 (1 h)			1 h, 250 GB			brain03, 2026-11-02
4,2	2520	651168	conforms	60.6	22	1 h, 250 GB	4833	5900122	brain03, 2026-11-02
4,3			🕓 (1 h)			1 h, 250 GB			brain03, 2026-11-02
5,2			🕓 (1 h)			1 h, 250 GB			brain03, 2026-11-02

stateless: pair fifo, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	sweeps	G_{ref}^{expl}	run
2,2	6	28	conforms	0.0	2	1 h, 250 GB	6	21	brain03, 2026-11-02
2,3	20	212	conforms	0.0	2	1 h, 250 GB	20	210	brain03, 2026-11-02
2,4	70	1676	conforms	0.2	2	1 h, 250 GB	77	2656	brain03, 2026-11-02
2,5	252	13604	conforms	1.1	2	48 h, 250 GB	510	61622	brain03, 2026-11-08
2,6	924	112380	conforms	9.5	2	1 h, 250 GB	2505	1201363	brain03, 2026-11-02
3,2	90	2898	conforms	0.3	2	1 h, 250 GB	129	5402	brain03, 2026-11-02
3,3	1680	330252	conforms	21.2	24	1 h, 250 GB	4560	3790449	brain03, 2026-11-02
3,4	34650	41133774	conforms	15370.0	204	48 h, 250 GB	79180	1333000700	brain02, 2026-11-08
4,2	2520	675960	conforms	60.2	22	1 h, 250 GB	8437	10077551	brain03, 2026-11-02
4,3			🕓 (1 h)			1 h, 250 GB			brain03, 2026-11-02
5,2			🕓 (1 h)			1 h, 250 GB			brain03, 2026-11-02

stateless: pair double-grant, model fifo, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	sweeps	G_{ref}^{expl}	run
2,2	6	70	64	violation	0.0	4	1 h, 250 GB	70	405	brain03, 2026-11-02
3,2	90	34650	34560	violation	31.8	38	1 h, 250 GB	34650	3114495	brain03, 2026-11-02

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	sweeps	G_{ref}^{expl}	run
3,3	1680			(1 h)			1 h, 250 GB			brain03, 2026-11-02

stateless: pair **never-grant**, model fifo, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	sweeps	G_{ref}^{expl}	run
2,2	6	21	8	violation	0.0	4	1 h, 250 GB	14	69	brain03, 2026-11-02
3,2	90	1504	716	violation	2.3	4	1 h, 250 GB	806	68535	brain03, 2026-11-02
3,3	1680	83852	40576	violation	346.1	36	1 h, 250 GB	42256	69579720	brain03, 2026-11-02

stateless: pair **wrong-round**, model fifo, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	sweeps	G_{ref}^{expl}	run
2,2	6	28	28	violation	0.0	4	1 h, 250 GB	28	168	brain03, 2026-11-02
3,2	90	2898	2898	violation	6.2	4	1 h, 250 GB	2898	260820	brain03, 2026-11-02
3,3	1680	330252	330252	violation	2820.5	98	1 h, 250 GB	330252	554823360	brain03, 2026-11-02

Not run (stateless). planned but never scheduled (grid points beyond the ladder): wrong-round 2 3 fifo, never-grant 2 3 fifo, double-grant 2 3 fifo, double-grant 2 4 fifo, never-grant 2 4 fifo, wrong-round 2 4 fifo, double-grant 2 5 fifo, wrong-round 2 5 fifo, never-grant 2 5 fifo, wrong-round 2 6 fifo, never-grant 2 6 fifo, double-grant 2 6 fifo, wrong-round 3 4 fifo, never-grant 3 4 fifo, wrong-round 3 4 fifo.t16.p.r3, control-lifo 3 5 fifo, fifo 3 5 fifo, wrong-round 3 5 fifo, never-grant 3 5 fifo, fifo 3 5 fifo.t16.p.r3, wrong-round 3 5 fifo.t16.p.r3, control-lifo 3 6 fifo, fifo 3 6 fifo, never-grant 3 6 fifo, … (94 instances in all).

left out by the table rules (ok): fifo 2 5 fifo.

left out by the table rules (timeout): fifo 3 4 fifo.

skipped: a smaller size had reached its limit: double-grant 3 4 fifo, double-grant 3 5 fifo, double-grant 3 6 fifo.

stateful

stateful: pair **control-lifo**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	run
2,2	6	28	conforms	0.0	4	1 h, 250 GB	brain02, 2026-11-02
2,3	20	212	conforms	0.0	2	1 h, 250 GB	brain02, 2026-11-02
2,4	70	1676	conforms	0.1	4	1 h, 250 GB	brain02, 2026-11-02
2,5	252	13604	conforms	0.3	4	1 h, 250 GB	brain02, 2026-11-02
2,6	924	112380	conforms	1.3	2	1 h, 250 GB	brain02, 2026-11-02
3,2	90	2850	conforms	0.1	4	1 h, 250 GB	brain02, 2026-11-02
3,3	1680	324282	conforms	2.6	4	1 h, 250 GB	brain02, 2026-11-02
3,4	34650	40400514	conforms	782.0	188	1 h, 250 GB	brain02, 2026-11-02
3,5			(1 h)			1 h, 250 GB	brain02, 2026-11-02
4,2	2520	651168	conforms	6.0	4	1 h, 250 GB	brain02, 2026-11-02
4,3			(1 h)			1 h, 250 GB	brain02, 2026-11-02
5,2			(1 h)			1 h, 250 GB	brain02, 2026-11-02

stateful: pair **fifo**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	run
2,2	6	28	conforms	0.0	2	1 h, 250 GB	brain02, 2026-11-02
2,3	20	212	conforms	0.0	4	1 h, 250 GB	brain02, 2026-11-02
2,4	70	1676	conforms	0.1	4	1 h, 250 GB	brain02, 2026-11-02
2,5	252	13604	conforms	0.7	2	48 h, 250 GB	brain03, 2026-11-08
2,6	924	112380	conforms	1.2	4	1 h, 250 GB	brain02, 2026-11-02
3,2	90	2898	conforms	0.1	4	1 h, 250 GB	brain02, 2026-11-02
3,3	1680	330252	conforms	2.8	4	1 h, 250 GB	brain02, 2026-11-02
3,4	34650	41133774	conforms	1303.4	178	48 h, 250 GB	brain02, 2026-11-08
3,5			(1 h)			1 h, 250 GB	brain02, 2026-11-02
4,2	2520	675960	conforms	6.1	6	1 h, 250 GB	brain02, 2026-11-02

size	G_{spec}	G_{impl}	verdict	T	M	tier	run
4,3			(1 h)			1 h, 250 GB	brain02, 2026-11-02
5,2			(1 h)			1 h, 250 GB	brain02, 2026-11-02

stateful: pair **double-grant**, model fifo, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	run
2,2	6	70	64	violation	0.0	4	1 h, 250 GB	brain02, 2026-11-02
2,3	20	924	904	violation	0.1	2	1 h, 250 GB	brain02, 2026-11-02
2,4	70	12870	12800	violation	0.5	28	1 h, 250 GB	brain02, 2026-11-02
2,5	252	184756	184504	violation	3.2	294	1 h, 250 GB	brain02, 2026-11-02
2,6	924	2704156	2703232	violation	42.5	4282	1 h, 250 GB	brain02, 2026-11-02
3,2	90	34650	34560	violation	0.6	48	1 h, 250 GB	brain02, 2026-11-02
3,3	1680	17153136	17151456	violation	171.9	15095	1 h, 250 GB	brain02, 2026-11-02
3,4				(1 h)			1 h, 250 GB	brain02, 2026-11-02
4,2	2520	63063000	63060480	violation	881.9	41278	1 h, 250 GB	brain02, 2026-11-02
4,3				(1 h)			1 h, 250 GB	brain02, 2026-11-02
5,2				(1 h)			1 h, 250 GB	brain02, 2026-11-02

stateful: pair **never-grant**, model fifo, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	run
2,2	6	21	8	violation	0.0	4	1 h, 250 GB	brain02, 2026-11-02
2,3	20	106	43	violation	0.0	4	1 h, 250 GB	brain02, 2026-11-02
2,4	70	545	224	violation	0.1	4	1 h, 250 GB	brain02, 2026-11-02
2,5	252	2864	1181	violation	0.2	4	1 h, 250 GB	brain02, 2026-11-02
2,6	924	15309	6320	violation	0.5	12	1 h, 250 GB	brain02, 2026-11-02
3,2	90	1504	716	violation	0.1	4	1 h, 250 GB	brain02, 2026-11-02
3,3	1680	83852	40576	violation	1.0	34	1 h, 250 GB	brain02, 2026-11-02
3,4	34650	5077092	2448528	violation	83.2	1062	1 h, 250 GB	brain02, 2026-11-02
3,5				(1 h)			1 h, 250 GB	brain02, 2026-11-02
4,2	2520	257844	136092	violation	2.9	164	1 h, 250 GB	brain02, 2026-11-02
4,3				(1 h)			1 h, 250 GB	brain02, 2026-11-02
5,2	113400	83803176	47273640	violation	3204.8	54125	1 h, 250 GB	brain02, 2026-11-02

stateful: pair **wrong-round**, model fifo, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	run
2,2	6	28	28	violation	0.0	4	1 h, 250 GB	brain02, 2026-11-02
2,3	20	212	212	violation	0.1	4	1 h, 250 GB	brain02, 2026-11-02
2,4	70	1676	1676	violation	0.2	4	1 h, 250 GB	brain02, 2026-11-02
2,5	252	13604	13604	violation	0.4	6	1 h, 250 GB	brain02, 2026-11-02
2,6	924	112380	112380	violation	1.7	14	1 h, 250 GB	brain02, 2026-11-02
3,2	90	2898	2898	violation	0.1	4	1 h, 250 GB	brain02, 2026-11-02
3,3	1680	330252	330252	violation	2.9	116	1 h, 250 GB	brain02, 2026-11-02
3,4	34650	41133774	41133774	violation	471.9	6435	48 h, 250 GB	brain04, 2026-11-04
3,5				(1 h)			1 h, 250 GB	brain02, 2026-11-02
4,2	2520	675960	675960	violation	7.2	602	1 h, 250 GB	brain02, 2026-11-02
4,3				(1 h)			1 h, 250 GB	brain02, 2026-11-02
5,2				(1 h)			1 h, 250 GB	brain02, 2026-11-02

Not run (stateful). planned but never scheduled (grid points beyond the ladder): fifo 3 4 fifo, fifo 3 5 fifo, wrong-round 3 5 fifo, fifo 3 5 fifo.t16.p.r3, wrong-round 3 5 fifo.t16.p.r3, wrong-round 3 6 fifo, never-grant 3 6 fifo, never-grant 4 4 fifo, double-grant 4 4 fifo, wrong-round 4 4 fifo, wrong-round 4 5 fifo, never-grant 4 5 fifo, never-grant 4 6 fifo, wrong-round 4 6 fifo, never-grant 5 3 fifo, wrong-round 5 3 fifo, double-grant 5 3 fifo, never-grant 5 4 fifo, wrong-round 5 4 fifo, double-grant 5 4 fifo, double-grant 5 5 fifo, wrong-round 5 5 fifo, never-grant 5 5 fifo, never-grant 5 6 fifo, … (43 instances in all).

left out by the table rules (ok): fifo 2 5 fifo, fifo 3 4 fifo, wrong-round 3 4 fifo.

skipped: a smaller size had reached its limit: double-grant 3 5 fifo, double-grant 3 6 fifo, control-lifo 3 6 fifo, fifo 3 6 fifo, fifo 4 4 fifo, double-grant 4 4 fifo, control-lifo 4 4 fifo, fifo 4 5 fifo, double-grant 4 5 fifo, control-lifo 4 5 fifo, fifo 4 6 fifo, double-grant 4 6 fifo, control-lifo 4 6 fifo, fifo 5 3 fifo, control-lifo 5 3 fifo, control-lifo 5 4 fifo, fifo 5 4 fifo, control-lifo 5 5 fifo, fifo 5 5 fifo, fifo 5 6 fifo, control-lifo 5 6 fifo, fifo 6 2 fifo, control-lifo 6 2 fifo, fifo 6 3 fifo, … (31 instances in all).

offline

offline: pair **control-lifo**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2,2	6	28	conforms	0.1	8	10 min, 16 GB	0.1	0.0	0.0	in-memory	brain01, 2026-11-02
3,2	90	2850	conforms	0.2	8	10 min, 16 GB	0.1	0.2	0.0	in-memory	brain01, 2026-11-02
3,3	1680	324282	conforms	3.8	34	1 h, 250 GB	0.3	3.1	0.7	merge	brain01, 2026-11-03

offline: pair **fifo**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2,2	6	28	conforms	0.1	8	10 min, 16 GB	0.1	0.0	0.0	in-memory	brain01, 2026-11-02
2,5	252	13604	conforms	0.4	8	48 h, 250 GB	0.1	0.4	0.0	merge	brain03, 2026-11-08
3,2	90	2898	conforms	0.2	8	10 min, 16 GB	0.1	0.2	0.0	in-memory	brain01, 2026-11-02
3,3	1680	330252	conforms	3.9	38	1 h, 250 GB	0.3	3.1	0.7	merge	brain01, 2026-11-03
3,4	34650	41133774	conforms	782.6	1344	48 h, 250 GB	4.6	654.2	128.4	merge	brain02, 2026-11-08

offline: pair **double-grant**, model fifo, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
2,2	6	70	28	violation	0.1	8	10 min, 16 GB	0.1	0.0	0.0	in-memory	brain01, 2026-11-02
3,2	90	34650	7044	violation	0.8	12	1 h, 250 GB	0.1	0.6	0.2	merge	brain01, 2026-11-03
3,3	1680	17153136	1823232	violation	253.9	4650	1 h, 250 GB	0.3	152.6	101.3	merge	brain01, 2026-11-03

offline: pair **never-grant**, model fifo, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
2,2	6	21	5	violation	0.1	8	10 min, 16 GB	0.1	0.1	0.0	in-memory	brain01, 2026-11-02
3,2	90	1504	176	violation	0.1	8	10 min, 16 GB	0.1	0.1	0.0	in-memory	brain01, 2026-11-02
3,3	1680	83852	4076	violation	1.3	76	10 min, 16 GB	0.3	0.9	0.4	in-memory	brain01, 2026-11-02

offline: pair **wrong-round**, model fifo, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
2,2	6	28	6	violation	0.1	8	10 min, 16 GB	0.1	0.0	0.0	in-memory	brain01, 2026-11-02
3,2	90	2898	372	violation	0.2	8	10 min, 16 GB	0.1	0.2	0.0	in-memory	brain01, 2026-11-02
3,3	1680	330252	13584	violation	3.7	147	10 min, 16 GB	0.3	2.9	0.7	in-memory	brain01, 2026-11-02
3,4	34650	41133774	542148	violation	739.4	1544	48 h, 250 GB	4.6	708.5	30.9	merge	brain02, 2026-11-08

Not run (offline). planned but never scheduled (grid points beyond the ladder): wrong-round 2 3 fifo, control-lifo 2 3 fifo, never-grant 2 3 fifo, double-grant 2 3 fifo, fifo 2 3 fifo, double-grant 2 4 fifo, fifo 2 4 fifo, control-lifo 2 4 fifo, never-grant 2 4 fifo, wrong-round 2 4 fifo, control-lifo 2 5 fifo, double-grant 2 5 fifo, fifo 2 5 fifo, wrong-round 2 5 fifo, never-grant 2 5 fifo, wrong-round 2 6 fifo, fifo 2 6 fifo, never-grant 2 6 fifo, control-lifo 2 6 fifo, double-grant 2 6 fifo, double-grant 3 4 fifo, control-lifo 3 4 fifo, wrong-round 3 4 fifo, never-grant 3 4 fifo, … (112 instances in all).

context-free twins offline, Rust implementation against the context-free Java port

context-free twins offline, Rust implementation against the context-free Java port: pair **control-lifo**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2,2	28	28	conforms	4.6	159	4 h, 250 GB	4.6	0.4	0.0	merge	brain02, 2026-11-03

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
3,2	2850	2850	conforms	8.6	470	10 min, 250 GB	8.6	0.3	0.0	merge	brain02, 2026-11-03
3,3	324282	324282	conforms	388.3	10552	10 min, 250 GB	387.2	3.8	1.1	merge	brain02, 2026-11-03

context-free twins offline, Rust implementation against the context-free Java port: pair **double-grant**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2,2	70	70	conforms	7.8	156	10 min, 250 GB	7.8	0.2	0.0	merge	brain02, 2026-11-03
3,2	34650	34650	conforms	102.5	6516	10 min, 250 GB	102.0	0.9	0.5	merge	brain02, 2026-11-03

context-free twins offline, Rust implementation against the context-free Java port: pair **fifo**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2,2	28	28	conforms	11.3	1814	4 h, 250 GB	11.3	0.1	0.0	merge	brain02, 2026-11-03
3,2	2898	2898	conforms	11.6	2784	10 min, 250 GB	11.4	0.3	0.2	merge	brain02, 2026-11-03
3,3	330252	330252	conforms	501.1	11680	10 min, 250 GB	499.7	3.5	1.4	merge	brain02, 2026-11-03

context-free twins offline, Rust implementation against the context-free Java port: pair **never-grant**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2,2	21	21	conforms	4.7	172	4 h, 250 GB	4.6	0.2	0.0	merge	brain02, 2026-11-03
3,2	1504	1504	conforms	8.1	366	10 min, 250 GB	8.1	0.2	0.0	merge	brain02, 2026-11-03
3,3	83852	83852	conforms	199.2	6920	10 min, 250 GB	198.2	1.2	1.0	merge	brain02, 2026-11-03

context-free twins offline, Rust implementation against the context-free Java port: pair **wrong-round**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2,2	28	28	conforms	4.4	175	4 h, 250 GB	4.4	0.2	0.0	merge	brain02, 2026-11-03
3,2	2898	2898	conforms	19.4	2134	10 min, 250 GB	19.4	0.4	0.0	merge	brain02, 2026-11-03
3,3	330252	330252	conforms	1018.1	12636	10 min, 250 GB	1016.9	3.8	1.2	merge	brain02, 2026-11-03

Not run (context-free twins offline, Rust implementation against the context-free Java port). planned but never scheduled (grid points beyond the ladder): control-lifo 3 4 fifo, never-grant 3 4 fifo, control-lifo 4 3 fifo, never-grant 4 3 fifo, wrong-round 4 3 fifo, fifo 4 3 fifo.

refused (an input missing at run time): double-grant 3 3 fifo.

context-free twins offline, Java port against the Rust implementation

context-free twins offline, Java port against the Rust implementation: pair **control-lifo**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2,2	28	28	conforms	4.6	159	4 h, 250 GB	0.4	4.6	0.0	merge	brain02, 2026-11-03
3,2	2850	2850	conforms	8.6	470	10 min, 250 GB	0.3	8.6	0.0	merge	brain02, 2026-11-03
3,3	324282	324282	conforms	388.5	10552	10 min, 250 GB	3.8	387.2	1.4	merge	brain02, 2026-11-03

context-free twins offline, Java port against the Rust implementation: pair **double-grant**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2,2	70	70	conforms	7.8	156	10 min, 250 GB	0.2	7.8	0.0	merge	brain02, 2026-11-03
3,2	34650	34650	conforms	102.5	6516	10 min, 250 GB	0.9	102.0	0.5	merge	brain02, 2026-11-03

context-free twins offline, Java port against the Rust implementation: pair **fifo**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2,2	28	28	conforms	11.3	1814	4 h, 250 GB	0.1	11.3	0.0	merge	brain02, 2026-11-03
3,2	2898	2898	conforms	11.4	2784	10 min, 250 GB	0.3	11.4	0.0	merge	brain02, 2026-11-03
3,3	330252	330252	conforms	501.0	11680	10 min, 250 GB	3.5	499.7	1.3	merge	brain02, 2026-11-03

context-free twins offline, Java port against the Rust implementation: pair **never-grant**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2,2	21	21	conforms	4.6	172	4 h, 250 GB	0.2	4.6	0.0	merge	brain02, 2026-11-03
3,2	1504	1504	conforms	8.1	366	10 min, 250 GB	0.2	8.1	0.0	merge	brain02, 2026-11-03
3,3	83852	83852	conforms	198.8	6920	10 min, 250 GB	1.2	198.2	0.6	merge	brain02, 2026-11-03

context-free twins offline, Java port against the Rust implementation: pair **wrong-round**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2,2	28	28	conforms	4.4	175	4 h, 250 GB	0.2	4.4	0.0	merge	brain02, 2026-11-03
3,2	2898	2898	conforms	19.4	2134	10 min, 250 GB	0.4	19.4	0.0	merge	brain02, 2026-11-03
3,3	330252	330252	conforms	1018.1	12636	10 min, 250 GB	3.8	1016.9	1.2	merge	brain02, 2026-11-03

Not run (context-free twins offline, Java port against the Rust implementation). planned but never scheduled (grid points beyond the ladder): control-lifo 3 4 fifo, never-grant 3 4 fifo, control-lifo 4 3 fifo, never-grant 4 3 fifo, wrong-round 4 3 fifo, fifo 4 3 fifo.

skipped: a smaller size had reached its limit: double-grant 3 3 fifo.

context-free twins online, Java port (JMC) against the Rust implementation's export

context-free twins online, Java port (JMC) against the Rust implementation's export: pair **control-lifo**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
2,2	28	28	conforms	4.0	164	1 h, 250 GB		brain02, 2026-11-03
3,2	2850	2850	conforms	10.6	1292	1 h, 250 GB		brain02, 2026-11-03
3,3	324282	324282	conforms	391.2	14271	1 h, 250 GB		brain02, 2026-11-03

context-free twins online, Java port (JMC) against the Rust implementation's export: pair **double-grant**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
2,2	70	70	conforms	5.8	175	1 h, 250 GB		brain02, 2026-11-03
3,2	34650	34650	conforms	62.4	7169	1 h, 250 GB		brain02, 2026-11-03

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
3,3	17153136		(1 h)			1 h, 250 GB		brain02, 2026-11-03

context-free twins online, Java port (JMC) against the Rust implementation's export: pair fifo, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
2,2	28	28	conforms	3.7	155	1 h, 250 GB		brain02, 2026-11-03
3,2	2898	2898	conforms	12.7	2959	1 h, 250 GB		brain02, 2026-11-03
3,3	330252	330252	conforms	489.1	13798	1 h, 250 GB		brain02, 2026-11-03

context-free twins online, Java port (JMC) against the Rust implementation's export: pair never-grant, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
2,2	21	21	conforms	3.7	158	1 h, 250 GB		brain02, 2026-11-03
3,2	1504	1504	conforms	11.0	2646	1 h, 250 GB		brain02, 2026-11-03
3,3	83852	83852	conforms	147.9	7630	1 h, 250 GB		brain02, 2026-11-03

context-free twins online, Java port (JMC) against the Rust implementation's export: pair wrong-round, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
2,2	28	28	conforms	3.7	166	1 h, 250 GB		brain02, 2026-11-03
3,2	2898	2898	conforms	10.9	3184	1 h, 250 GB		brain02, 2026-11-03
3,3	330252	330252	conforms	398.8	13563	1 h, 250 GB		brain02, 2026-11-03

context-free twins online, Rust implementation (TraceForge) against the Java port's export

context-free twins online, Rust implementation (TraceForge) against the Java port's export: pair control-lifo, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
2,2	28	28	conforms	4.1	168	1 h, 250 GB		brain02, 2026-11-03
3,2	2850	2850	conforms	9.4	732	1 h, 250 GB		brain02, 2026-11-03
3,3	324282	324282	conforms	605.5	16557	1 h, 250 GB		brain02, 2026-11-03

context-free twins online, Rust implementation (TraceForge) against the Java port's export: pair double-grant, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
2,2	70	70	conforms	3.6	154	1 h, 250 GB		brain02, 2026-11-03
3,2	34650	34650	conforms	41.6	8217	1 h, 250 GB		brain02, 2026-11-03

context-free twins online, Rust implementation (TraceForge) against the Java port's export: pair fifo, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
2,2	28	28	conforms	4.3	150	1 h, 250 GB		brain02, 2026-11-03
3,2	2898	2898	conforms	10.7	3838	1 h, 250 GB		brain02, 2026-11-03

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
3,3	330252	330252	conforms	676.0	18371	1 h, 250 GB		brain02, 2026-11-03

context-free twins online, Rust implementation (TraceForge) against the Java port's export: pair **never-grant**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
2,2	21	21	conforms	5.1	154	1 h, 250 GB		brain02, 2026-11-03
3,2	1504	1504	conforms	9.7	610	1 h, 250 GB		brain02, 2026-11-03
3,3	83852	83852	conforms	208.7	10306	1 h, 250 GB		brain02, 2026-11-03

context-free twins online, Rust implementation (TraceForge) against the Java port's export: pair **wrong-round**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
2,2	28	28	conforms	4.2	164	1 h, 250 GB		brain02, 2026-11-03
3,2	2898	2898	conforms	9.8	761	1 h, 250 GB		brain02, 2026-11-03
3,3	330252	330252	conforms	621.5	17801	1 h, 250 GB		brain02, 2026-11-03

Not run (context-free twins online, Rust implementation (TraceForge) against the Java port's export).
skipped: a smaller size had reached its limit: double-grant 3 3 fifo.